

LAB EXPERIMENT RECORD

SET 5: WEBSITE ANALYTICS

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Aim

To visualize website traffic data using a line chart, a bar chart and a stacked area chart in R, and to build an interactive dashboard using Tableau.

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

Additional columns used: The given table contains only Date, Page Views and Click-through Rate. For Program 3, the number of user interactions (likes, shares and comments) was added for the same five dates:

Date	Likes	Shares	Comments
2023-01-01	120	40	25
2023-01-02	150	55	30
2023-01-03	110	35	20
2023-01-04	170	60	35
2023-01-05	190	70	45

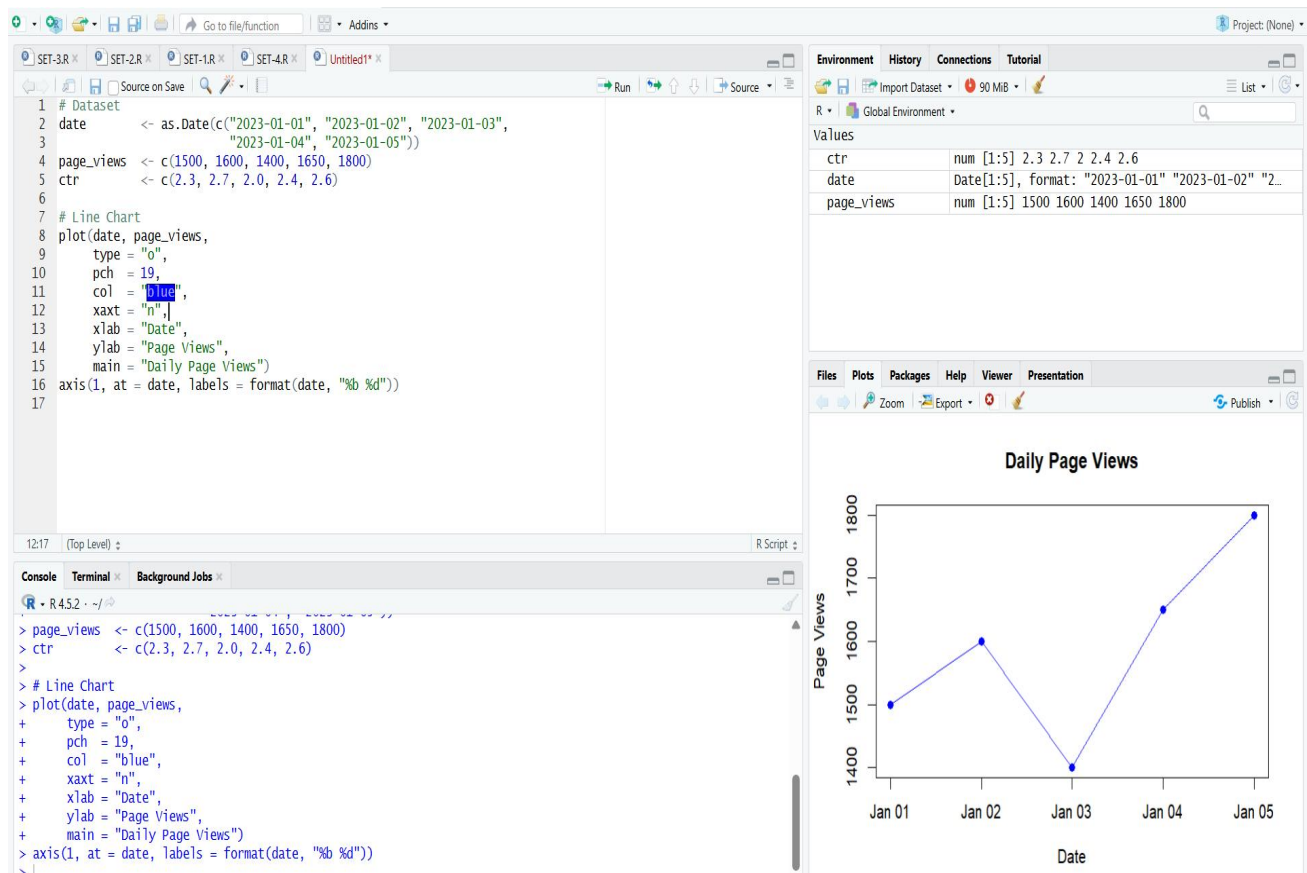
PROGRAM 1: Line Chart of Daily Page Views

Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
# Dataset
date      <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                      "2023-01-04", "2023-01-05"))
page_views <- c(1500, 1600, 1400, 1650, 1800)
ctr        <- c(2.3, 2.7, 2.0, 2.4, 2.6)

# Line Chart
plot(date, page_views,
     type = "o",
     pch = 19,
     col = "blue",
     xaxt = "n",
     xlab = "Date",
     ylab = "Page Views",
     main = "Daily Page Views")
axis(1, at = date, labels = format(date, "%b %d"))
```

OUTPUT:



Output 1: Daily Page Views (Line Chart)

Observation: Page views rose from 1,500 on 1 January to 1,600 on 2 January, dropped to the lowest value of 1,400 on 3 January, and then increased to 1,650 and finally to the highest value of 1,800 on 5 January. Overall, traffic shows an upward trend of 20% across the five days, with one dip on 3 January.

PROGRAM 2: Bar Chart of Top N Days by Click-through Rate

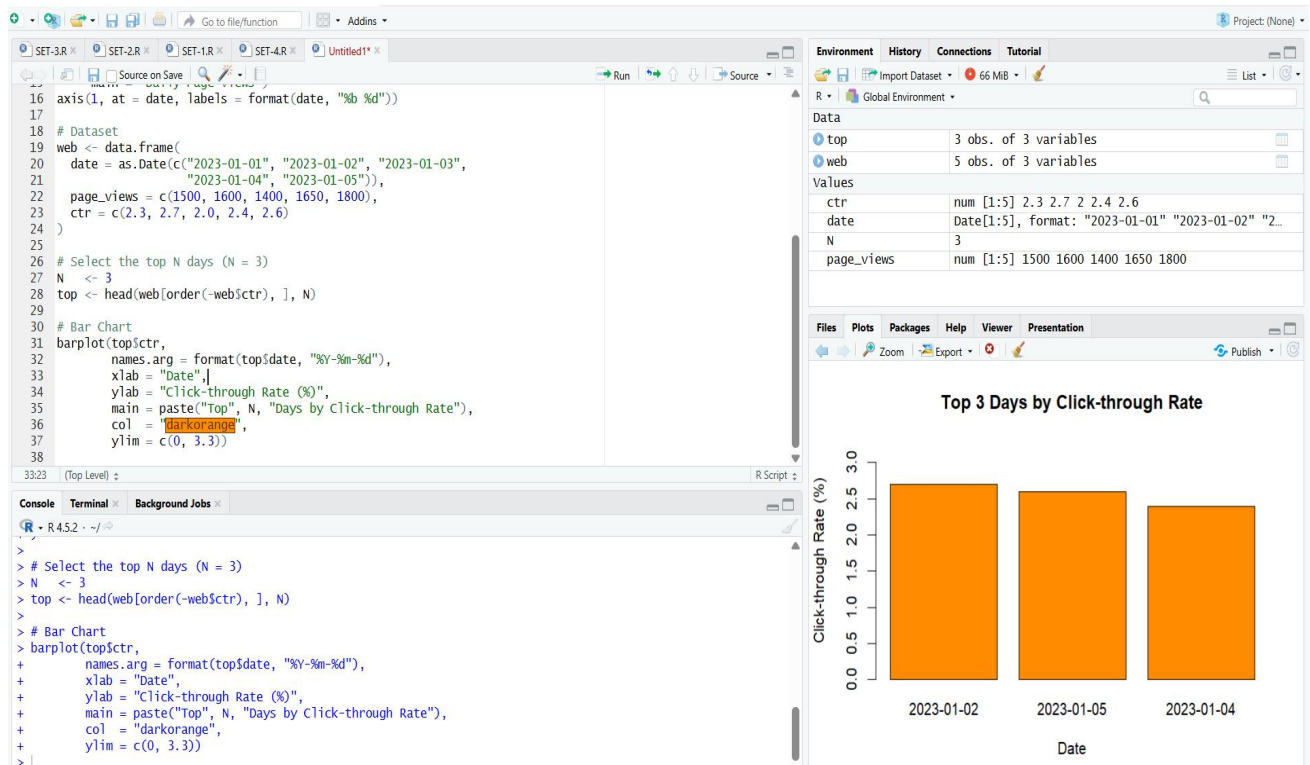
Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
# Dataset
web <- data.frame(
  date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                  "2023-01-04", "2023-01-05")),
  page_views = c(1500, 1600, 1400, 1650, 1800),
  ctr = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

# Select the top N days (N = 3)
N <- 3
top <- head(web[order(-web$ctr), ], N)

# Bar Chart
barplot(top$ctr,
        names.arg = format(top$date, "%Y-%m-%d"),
        xlab = "Date",
        ylab = "Click-through Rate (%)",
        main = paste("Top", N, "Days by Click-through Rate"),
        col = "darkorange",
        ylim = c(0, 3.3))
```

OUTPUT:



Output 2: Top 3 Days by Click-through Rate (Bar Chart)

Observation: The three days with the highest click-through rate are 2 January (2.7%), 5 January (2.6%) and 4 January (2.4%). The lowest rate, 2.0% on 3 January, is left out of the top 3. The value of N can be changed in the code to show more or fewer days. The average click-through rate over all five days is 2.4%.

PROGRAM 3: Stacked Area Chart of User Interactions

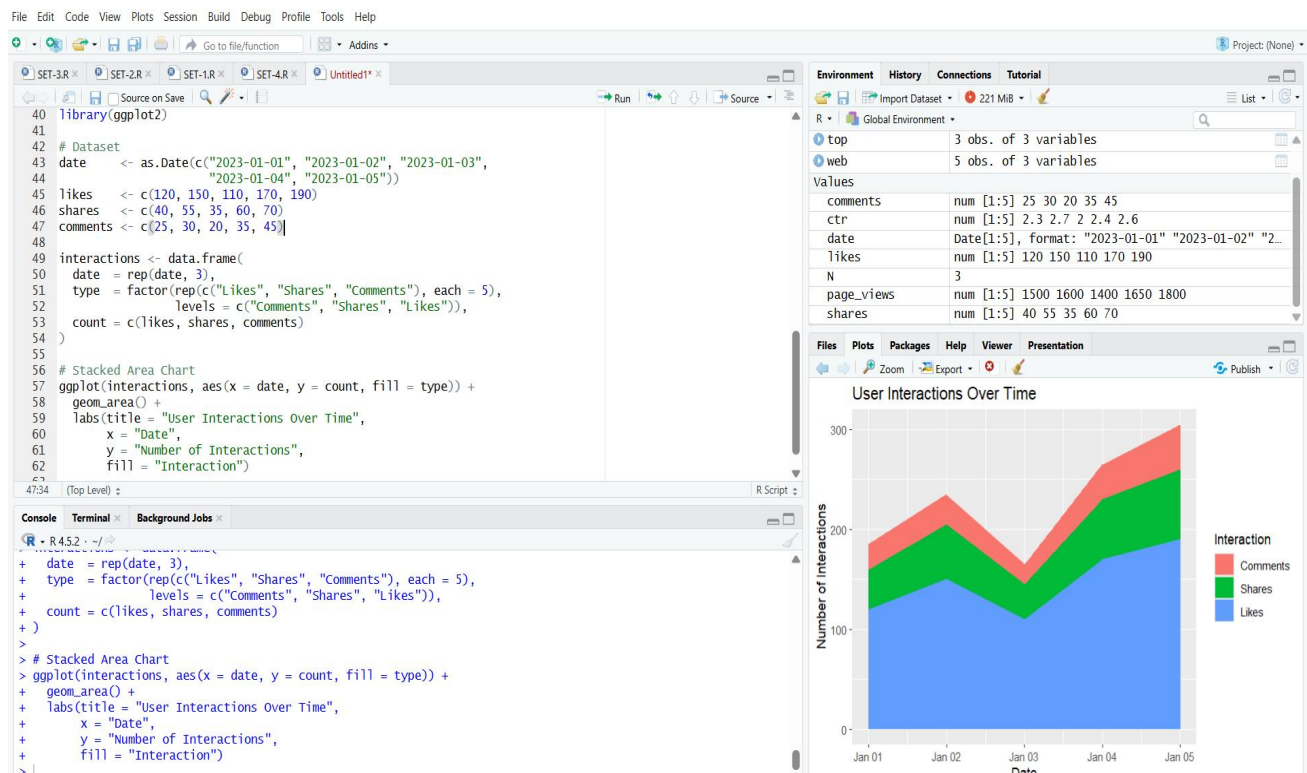
Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

```
library(ggplot2)

# Dataset
date      <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                       "2023-01-04", "2023-01-05"))
likes     <- c(120, 150, 110, 170, 190)
shares    <- c(40, 55, 35, 60, 70)
comments  <- c(25, 30, 20, 35, 45)
interactions <- data.frame(
  date = rep(date, 3),
  type = factor(rep(c("Likes", "Shares", "Comments"), each = 5),
                levels = c("Comments", "Shares", "Likes")),
  count = c(likes, shares, comments)
)

# Stacked Area Chart
ggplot(interactions, aes(x = date, y = count, fill = type)) +
  geom_area() +
  labs(title = "User Interactions Over Time",
       x = "Date",
       y = "Number of Interactions",
       fill = "Interaction")
```

OUTPUT:



Output 3: User Interactions Over Time (Stacked Area Chart)

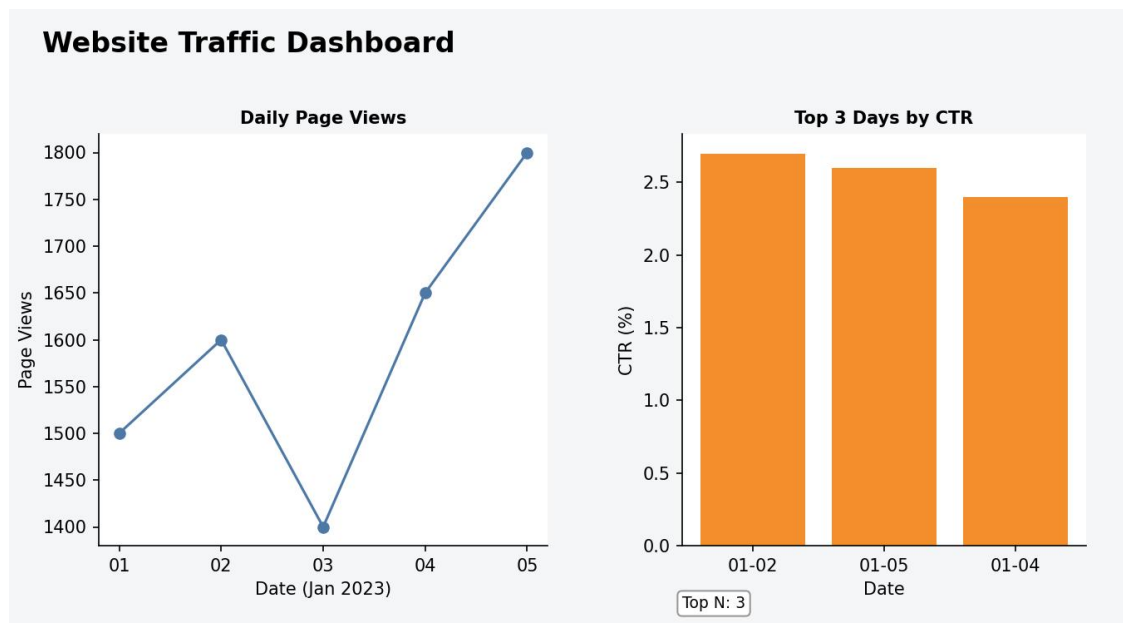
Insight: Likes make up the largest part of the interactions (740 of 1,155, about 64%), followed by shares (260) and comments (155). Total interactions follow the same pattern as page views: they dip to the lowest value (165) on 3 January and reach the highest value (305) on 5 January. The three layers rise and fall together, so a busy day increases all types of interaction.

PROGRAM 4: Interactive Website Traffic Dashboard in Tableau

Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

Steps in Tableau

1. Save the website traffic data (Date, Page Views, Click-through Rate) as an Excel or CSV file. Open Tableau and connect to the file using Connect > Microsoft Excel / Text file. Make sure Date has the Date data type.
2. Sheet 1 (Line Chart): Drag Date to Columns (set to Day, continuous) and Page Views to Rows (SUM). Add Page Views to Label and rename the sheet "Daily Page Views".
3. Create a parameter named "Top N" (Integer, value 3, range 1 to 5) and show the parameter control.
4. Sheet 2 (Bar Chart): Drag Date (Day, discrete) to Columns and Click-through Rate to Rows (AVG). Right-click Date > Filter > Top, choose "By field", and set the value to the Top N parameter by Click-through Rate. Sort the bars in descending order and rename the sheet "Top N Days by CTR".
5. Create a new Dashboard (Dashboard > New Dashboard). Set the size to Automatic and drag both sheets onto the dashboard area.
6. Add interactivity: select each chart and click Use as Filter, so selecting a date in one chart filters the other chart. Add a Date range filter (Show Filter > Apply to Worksheets > Selected Worksheets).
7. Add the title "Website Traffic Dashboard", keep the Top N control visible, and test the dashboard in Presentation Mode.



Output 4: Interactive Website Traffic Dashboard (Tableau)

Observation: The dashboard shows the page view trend and the top days by click-through rate side by side. Users can change the Top N value, select a date range, or click a point or bar to see how page views and click-through rate relate for a particular day.

Procedure

1. Define the date, page views and click-through rate vectors in R, converting the dates with `as.Date()`.
2. Use the `plot()` function with `type = "o"` to create the line chart of daily page views, and `axis()` to label the dates.

3. Sort the data with `order()` and select the top N rows using `head()`, then use `barplot()` to create the bar chart.
4. Create a long-format data frame of likes, shares and comments, and use `ggplot2` with `geom_area()` to draw the stacked area chart.
5. Load the website traffic file into Tableau and build the line chart and top N bar chart worksheets.
6. Combine both worksheets in a Tableau dashboard and add the Top N parameter, filters and actions for interactivity.

RESULT:

The website traffic data was successfully visualized using a line chart, a bar chart and a stacked area chart in R. An interactive Tableau dashboard was also created, combining the line chart and the top N bar chart so that users can explore page views and click-through rates.

Key Functions and Tools Used

Function / Tool	Purpose
<code>as.Date()</code>	Converts text into date values.
<code>plot()</code> / <code>axis()</code>	Creates the line chart and adds custom date labels.
<code>order()</code> / <code>head()</code>	Sorts the data and selects the top N days.
<code>barplot()</code>	Creates the bar chart.
<code>ggplot()</code> / <code>geom_area()</code>	Creates the stacked area chart.
Tableau Parameter	Lets the user choose the value of N.
Tableau Dashboard	Combines worksheets with filters and actions.